



Modelling Financial Markets

Chapter 3: Factor Models and Asset Pricing



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Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ APT: Arbitrage Pricing Theory
- ▣ BAB: Betting Against Beta
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BH: Benjamini–Hochberg (procedure)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CAGR: Compound Annual Growth Rate
- ▣ CAPM: Capital Asset Pricing Model
- ▣ CMA: Conservative Minus Aggressive (investment factor)
- ▣ CML: Capital Market Line
- ▣ ETF: Exchange-Traded Fund
- ▣ FDR: False Discovery Rate
- ▣ FF3: Fama–French three-factor model
- ▣ FF5: Fama–French five-factor model
- ▣ FWER: Family-Wise Error Rate
- ▣ GLS: Generalised Least Squares
- ▣ GMM: Generalized Method of Moments
- ▣ GRS: Gibbons–Ross–Shanken (test)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HML: High Minus Low (value factor)
- ▣ IPCA: Instrumented Principal Component Analysis
- ▣ LASSO: Least Absolute Shrinkage and Selection Operator
- ▣ MOM: MOMentum factor
- ▣ NYSE: New York Stock Exchange
- ▣ OLS: Ordinary Least Squares
- ▣ OMV: Österreichische Mineralölverwaltung (Austrian Mineral Oil Administration (OMV group))

Acronyms used in this material (2/2)

- ▣ PC1: first Principal Component
- ▣ PC2: second Principal Component
- ▣ PC3: third Principal Component
- ▣ PCA: Principal Component Analysis
- ▣ PRDS: Positive Regression Dependence on a Subset (of p-values)
- ▣ RF: Risk-Free rate
- ▣ RMSE: Root Mean Squared Error
- ▣ RMW: Robust Minus Weak (profitability factor)
- ▣ ROE: Return On Equity
- ▣ RP-PCA: Risk-Premium Principal Component Analysis
- ▣ SDF: Stochastic Discount Factor
- ▣ SMB: Small Minus Big (size factor)
- ▣ SML: Security Market Line
- ▣ UMD: Up Minus Down (momentum factor)
- ▣ US: United States

Chapter Route and Outcomes

Four questions

- ▣ What explains expected returns? The CAPM and multifactor models
- ▣ How are exposures estimated? Betas, their uncertainty and shrinkage
- ▣ How are pricing restrictions tested? GRS, Fama–MacBeth, model comparison
- ▣ Do the findings survive? Multiple testing, latent factors, replication

After this chapter you can

- ▣ estimate a beta with HAC errors and shrink it
- ▣ test the CAPM jointly and estimate risk premia with robust errors
- ▣ judge a new factor against multiple-testing hurdles and replication evidence

Part I: Sections 1–5; Part II: Sections 6–13; technical detail in the [Appendix](#)

What Is a Factor Model?

- A **factor model** explains the return of asset i by a few common variables $f_{1,t}, \dots, f_{K,t}$, the **factors**

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_{i,1}f_{1,t} + \dots + \beta_{i,K}f_{K,t} + \varepsilon_{i,t}$$

- $\beta_{i,k}$: **factor loading** (exposure) of asset i to factor k ; $\varepsilon_{i,t}$: asset-specific (idiosyncratic) shock
- Two uses in this chapter
 - ▶ Explain an asset: the technology ETF XLK moves 1.28 times the US market (Section 3)
 - ▶ Evaluate a fund: is its α_i positive once its exposures are priced (Sections 4 and 10)?
- Also: cost of capital, risk management (N assets need $N(N+1)/2$ covariances, K factors far fewer), portfolio construction (Campbell, Lo & MacKinlay (1997), ch. 5–6)
- Key message: expected returns should be explained by **exposures to priced risks**, not by the name of the asset

Data and Conventions

Series	Content	Period
Fama–French factors	market, size, value, profitability, investment	1963–2026
Momentum factor	winners minus losers	1963–2026
25 size $\times$ B/M portfolios	US stocks, two sorts	1963–2026
US sector ETFs	S&P 500 sectors: 9 from 1999, 11 from 2018	1999–2026
Factor ETFs	momentum, value, quality, min. volatility, size	2013–2026
BVB blue chips, BET	11 stocks and their index	2015–2026

- ▣ Factors and portfolios: [Kenneth French Data Library](#), in percent (divide by 100); ETFs and stocks: daily adjusted closing prices (BET: closing values), to 18 Sep 2026
- ▣ **Excess return** $R_{i,t}^e = R_{i,t} - R_{f,t}$; $R_{f,t}$: one-month Treasury bill rate (RF in the French data)
- ▣ **B/M**: book value of equity over its market value; **BET**: the benchmark index of the BVB
- ▣ Simple returns for regressions on the French factors; log returns for rolling, BVB and PCA analyses
- ▣ BVB: daily $R_{f,t}$ treated as constant; it cancels in covariances, so betas are unchanged
- ▣ ETFs and stocks: dividend- and split-adjusted prices; indices: closing values

Mean-Variance and the Tangency Portfolio

- **Markowitz (1952)**: with weights w , expected excess returns μ and covariance Σ , the frontier solves

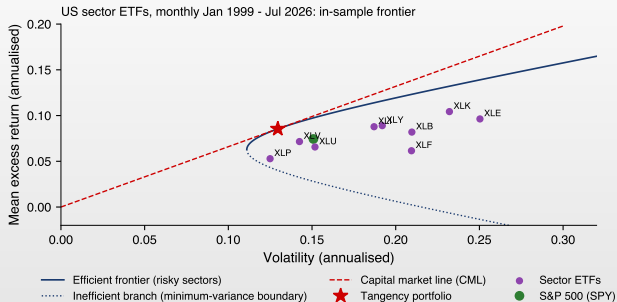
$$\min_w w^\top \Sigma w \quad \text{s.t.} \quad w^\top \mu = \mu^*, \quad w^\top \mathbf{1} = 1$$

- **Sharpe ratio** $SR_p = \mathbb{E}[R_p - R_f] / \sigma_p$; with a risk-free asset the best risky portfolio is the **tangency portfolio**

$$w_{\text{tan}} = \Sigma^{-1} \mu / \mathbf{1}^\top \Sigma^{-1} \mu$$

- **CML**: mixes of R_f and the tangency portfolio, $\mu_p = SR_{\text{tan}} \sigma_p$; negative weights are **short positions**
- From the tangency portfolio to the beta pricing equation: Seminar 3, problem A1; estimation error and shrinkage: [Chapter 4](#)

The Efficient Frontier of US Sectors



- Monthly excess returns of 9 sector ETFs, January 1999 – July 2026 ($T = 331$ months)
- Tangency portfolio: mean 8.55%, volatility 12.96%, Sharpe ratio 0.66
- SPY (the S&P 500 ETF, the market proxy): mean 7.44%, volatility 15.08%, Sharpe ratio 0.49: here the market is **not** on the frontier, as the CAPM would require
- Tangency weights: -0.37 in financials, -0.21 in materials, $+0.37$ in health care
- The frontier is **in-sample**; estimated means are very noisy ([Chapter 4](#))

MFM_ch3_frontier

The CAPM: Sharpe, Lintner, Black



William F. Sharpe, Nobel Prize in
Economics 1990

Photo: Larry D. Moore (2007); CC BY
4.0; Wikimedia Commons

- **CAPM**: Capital Asset Pricing Model, [Sharpe \(1964\)](#) and [Lintner \(1965\)](#)
- If every investor holds a mean–variance efficient portfolio, all hold the same tangency portfolio, which must then be the **market portfolio**
- Expected excess return is proportional to **beta**

$$\mathbb{E}[R_i] - R_f = \beta_i (\mathbb{E}[R_m] - R_f), \quad \beta_i = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)}$$

- $\mathbb{E}[R_m] - R_f$: **market risk premium**; β_i : sensitivity of asset i to the market
- [Black \(1972\)](#): a version without a risk-free asset, the zero-beta CAPM (Section 4)
- Sharpe shared the 1990 Nobel Prize with Markowitz and Miller

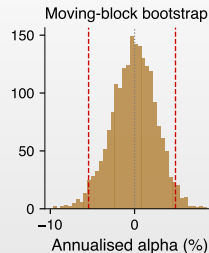
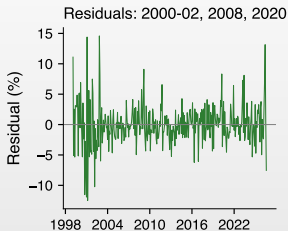
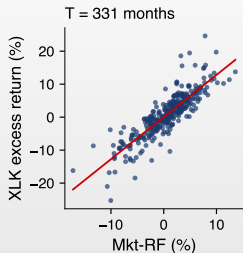
What the CAPM Implies

- Main assumptions
 - ▶ Mean–variance investors with identical beliefs, one period
 - ▶ Unlimited borrowing and lending at R_f ; no taxes or trading costs
- Only beta is priced: no other characteristic should predict returns; idiosyncratic risk earns nothing
- **SML** (Security Market Line): every asset lies on the line (beta, mean) with slope equal to the premium; the CML contains only efficient portfolios. The SML is the testable statement (Section 4)
- Cost of capital from the CAPM needs an estimated beta and an estimated premium: the error of the premium dominates
- Which portfolio is “the market”? The answer decides what a test of the CAPM can establish (Roll’s critique, Section 4)

The Market Model, HAC Errors and Estimation Choices

- ▣ Market model $R_{i,t}^e = \alpha_i + \beta_i R_{m,t}^e + \varepsilon_{i,t}$; **OLS** (Ordinary Least Squares): $\hat{\beta}_i = \widehat{\text{Cov}}(R_i^e, R_m^e) / \widehat{\text{Var}}(R_m^e)$
- ▣ Variance split $\sigma_i^2 = \beta_i^2 \sigma_m^2 + \sigma_{\varepsilon,i}^2$: systematic (priced under the CAPM) plus idiosyncratic (diversifiable); R^2 is the systematic share
- ▣ Volatility clustering ([Chapter 1](#)) makes residuals heteroskedastic: use **HAC** errors, [Newey & West \(1987\)](#), with Bartlett weights and $L = \lfloor 4(T/100)^{2/9} \rfloor$ lags ($T = 331$: $L = 5$; $T = 757$: $L = 6$); the point estimates are those of OLS
- ▣ Choices to report with every beta
 - ▶ Frequency and window: 3–5 years of monthly or 1–2 years of daily data; daily data suffer from asynchronous trading
 - ▶ Market proxy: SPY (S&P 500) for US assets, BET for Romanian stocks
 - ▶ Data cleaning: an unadjusted corporate event creates a fake return (DIGI, May 2017: a daily log return of -6.6)

Worked Example: Beta of the Technology ETF

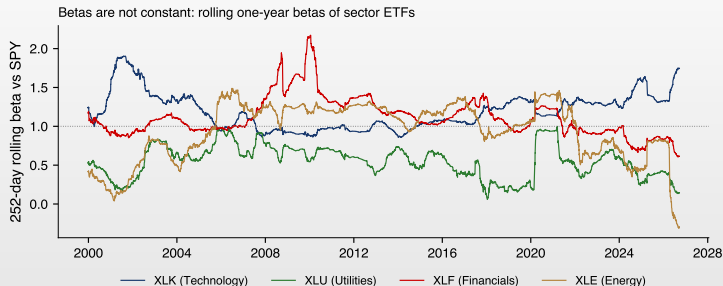


- Monthly excess returns
- OLS: beta = 1.277
- OLS residual
- Bootstrap alpha (2000 draws)
- - - 95% percentile interval

	Estimate	Classical	White	Newey–West
α (ann.)	0.33%	2.27%	2.32%	2.45%
β	1.277	0.041	0.054	0.082

- XLK on Mkt-RF, monthly 1999–2026, $T = 331$, $R^2 = 0.742$
- Robust s.e. of β : twice the classical one; heteroskedasticity explains part (White), serial correlation of $f_t \varepsilon_t$ the rest
- $t(\alpha) = 0.33/2.45 = 0.13$; bootstrap interval $[-5.5\%, 4.9\%]$

Betas Are Not Constant



- OLS beta on the last 252 daily log returns, recomputed every day
- Financials (XLF): from 0.60 to 2.17; the peak follows the 2008–2009 banking crisis
- Technology (XLK): from 0.86 to 1.90; high in the dot-com years, 1.75 at the end
- Energy (XLE): from -0.31 to 1.49; negative at the end, when oil moved against the market
- Utilities (XLU): from 0.06 to 1.00, the classic defensive sector
- A full-sample beta hides these regimes; conditional models: [Chapters 5–6](#)

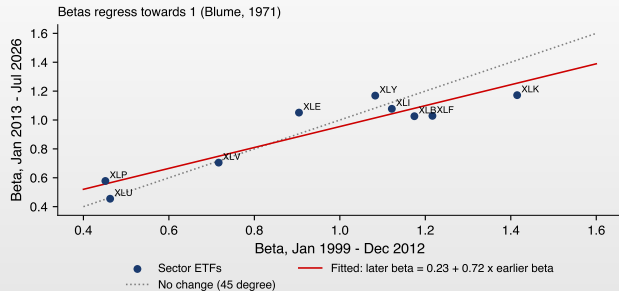
Shrinking Betas: Blume and Vasicek

- ▣ **Regression to the mean:** extreme estimated betas are partly noise and move towards 1 later
- ▣ **Blume (1971):** regress later betas on earlier betas; the practitioner rule is $\beta^{\text{Blume}} = 0.33 + 0.67 \hat{\beta}$
- ▣ **Vasicek (1973):** Bayesian average of the estimate and the cross-sectional prior mean $\bar{\beta}$

$$\beta_i^V = w_i \hat{\beta}_i + (1 - w_i) \bar{\beta}, \quad w_i = \frac{\sigma_{\beta}^2}{\sigma_{\beta}^2 + \text{se}(\hat{\beta}_i)^2}$$

- ▣ σ_{β}^2 is the variance of the **true** betas: $\text{Var}_{cs}(\hat{\beta}) = \sigma_{\beta}^2 + \overline{\text{se}^2}$, so the empirical-Bayes prior is $\sigma_{\beta}^2 = \text{Var}_{cs}(\hat{\beta}) - \overline{\text{se}^2}$ (posterior derivation: Seminar 3, problem A3)
- ▣ XLK, monthly 1999–2012, 9 sectors: $\hat{\beta} = 1.415$ (se 0.139), $\bar{\beta} = 0.949$, $\sigma_{\beta}^2 = 0.116 - 0.008 = 0.108$, $w = 0.849$
 ► $\beta^V = 0.849 \times 1.415 + 0.151 \times 0.949 = 1.344$; Blume 1.278; beta estimated on 2013–2026: 1.172

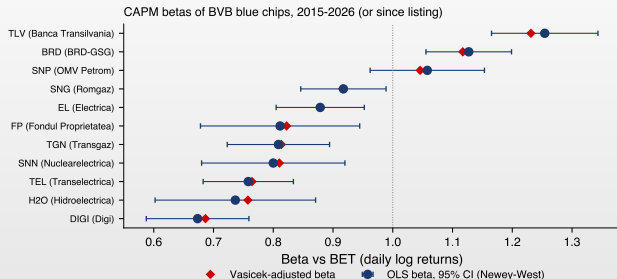
Do Betas Regress Towards One?



- Monthly betas of 9 sector ETFs on SPY: 1999–2012 (horizontal) vs 2013–2026 (vertical)
- Fitted line $\hat{\beta}_{\text{later}} = 0.23 + 0.72 \hat{\beta}_{\text{earlier}}$, very close to Blume's $0.33 + 0.67 \hat{\beta}$
- RMSE of the forecast of later betas, $\sqrt{\frac{1}{N} \sum_i (\text{forecast}_i - \hat{\beta}_{i,\text{later}})^2}$: raw 0.135, Blume 0.109, Vasicek 0.114
- Shrinkage cuts the forecast error by 15–20%; with 9 sectors the lesson is qualitative

 MFM_ch3_beta_shrink

CAPM Betas on the Bucharest Stock Exchange



- Daily log returns on BET, from 6 January 2015 or the listing date (Digi, DIGI, 2017; Hidroelectrica, H2O, 2023); days with $|r_t| > 0.5$ removed (one for DIGI, two for Fondul Proprietatea, FP)
- Banks are the most market-sensitive: Banca Transilvania (TLV) 1.25, BRD 1.13; OMV Petrom 1.06; Transelectrica 0.76, Digi 0.67
- R^2 : 0.59 for TLV, only 0.21 for DIGI; large index weights raise R^2 mechanically
- Vasicek changes little: the daily estimates are precise (se ≈ 0.04 – 0.07); stale prices bias daily betas of less liquid stocks (Seminar 3, problem B4)

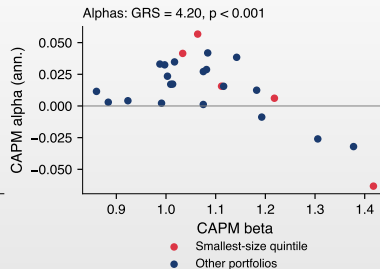
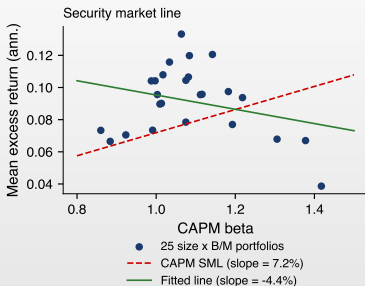
From One Alpha to the GRS Test

- ▣ **Jensen's alpha** (Jensen (1968)) is the market-model intercept α_i ; the CAPM implies $\alpha_i = 0$ for every asset
- ▣ One t -test per asset (XLK: $t = 0.13$) inflates false rejections across many assets (Section 8): **GRS**, Gibbons, Ross & Shanken (1989), tests $H_0 : \alpha_1 = \dots = \alpha_N = 0$ jointly

$$W = \frac{T - N - 1}{N} \frac{\hat{\alpha}^\top \hat{\Sigma}^{-1} \hat{\alpha}}{1 + \hat{\mu}_m^2 / \hat{\sigma}_m^2} \sim F(N, T - N - 1)$$

- ▣ $\hat{\Sigma}$: maximum-likelihood residual covariance (divided by T); $\hat{\mu}_m / \hat{\sigma}_m$: Sharpe ratio of the market
- ▣ Exact, conditional on the factors, under i.i.d. multivariate Normal errors independent of the factors, with zero mean and constant nonsingular covariance; F : Fisher distribution
- ▣ Large W : the market is far from the mean–variance frontier of the test assets

The Security Market Line in the Data



- Test assets: US stocks sorted each June into 5 size and 5 B/M quintiles, 25 value-weighted portfolios (Fama & French (1993)), July 1963 – July 2026, $T = 757$
- The CAPM line has slope 7.2% a year; the fitted line has slope -4.4%: higher beta, **lower** average return

- Betas span only 0.86–1.42, average excess returns 3.9%–13.3%
- Small-value alpha +5.7%, small-growth -6.3% a year; 10 of 25 alphas have $|t| > 1.96$
- GRS = 4.20, p -value 8×10^{-11} : the CAPM is rejected

GRS as a Sharpe-Ratio Test

- Identity (Gibbons, Ross & Shanken (1989)): with $\hat{\Sigma}$ and the moments estimated by maximum likelihood

$$\hat{\alpha}^\top \hat{\Sigma}^{-1} \hat{\alpha} = \widehat{SR}^2(f, R) - \widehat{SR}^2(f)$$

- $\widehat{SR}^2(f, R)$: squared Sharpe ratio of the tangency portfolio of the factor and the N test assets

- So GRS measures the Sharpe-ratio gain from the test assets

$$W = \frac{T - N - 1}{N} \cdot \frac{\widehat{SR}^2(f, R) - \widehat{SR}^2(f)}{1 + \widehat{SR}^2(f)}$$

- 25 portfolios, monthly Sharpe ratios: market 0.134 (0.47 a year); market plus portfolios 0.405 (1.40 a year)
- $W = 29.24 \times (0.405^2 - 0.134^2)/(1 + 0.134^2) = 4.20$; the gain expected under H_0 is far smaller ([Appendix](#))

When the GRS Assumptions Fail

- ▣ **Heavy tails and volatility clustering** (Chapter 1): the F distribution is no longer exact
 - ▶ Residual bootstrap under H_0 : resample whole rows of residuals, rebuild returns with $\alpha = 0$ (Seminar 3, problem B11); robust to non-Normality only if the errors are i.i.d. and independent of the factors
 - ▶ GMM Wald test: moments $\varepsilon_t \otimes (1, f_t)$, $W_{\text{GMM}} = \hat{\alpha}^\top \hat{V}_\alpha^{-1} \hat{\alpha} \rightarrow \chi_N^2$, with $\hat{V}_\alpha$ the HAC estimate of $\text{Var}(\hat{\alpha})$: robust to heteroskedasticity and autocorrelation
- ▣ **Rank condition**: $\hat{\Sigma}$ has rank at most $T - 2$, so GRS is not computable when $N \geq T - 1$, and its power falls well before that point
- ▣ Tests built for many assets: [▶ Appendix](#)

A Flat SML: Black and Betting Against Beta



Fischer Black (1938–1995)

Photo: Dalmatine at English Wikipedia
(2009 upload); public domain; Wikimedia
Commons

- **Black (1972)**: if borrowing is restricted, investors who want more risk buy high-beta assets instead of using leverage; high-beta assets become expensive and earn less
- Prediction: the SML is **flatter** than the CAPM line and starts at a **zero-beta** rate above R_f
- **Frazzini & Pedersen (2014)** turn this into a strategy, **BAB** (Betting Against Beta)

$$r_t^{\text{BAB}} = \frac{R_{L,t} - R_{f,t}}{\beta_L} - \frac{R_{H,t} - R_{f,t}}{\beta_H}$$

- L : low-beta portfolio leveraged to beta 1; H : high-beta portfolio de-leveraged to beta 1; BAB has beta zero by construction and a positive average return in their data
- An unscaled low-minus-high-beta spread (Seminar 3, problem C1) is not beta-neutral: its mean mixes the premium with a negative market exposure

What a CAPM Test Establishes

- ▣ Roll (1977): the CAPM is a statement about the **true** market portfolio of all wealth; stock indices omit bonds, real estate, private firms and human capital
- ▣ Any efficient portfolio prices all assets exactly by beta: with R_f on the CML, with a zero-beta rate (Black (1972)) on the risky-asset frontier (Seminar 3, problem A1)
 - ▶ A CAPM test is a joint test of the model and of the efficiency of the proxy
- ▣ **Test assets matter**: 9 sector ETFs, January 1999 – July 2026 ($T = 331$): $GRS = 0.53$, $p = 0.85$, not rejected
 - ▶ On the same 331 months the 25 size $\times$ B/M portfolios still reject: $GRS = 2.91$, $p = 9 \times 10^{-6}$
 - ▶ Sectors mix small and big, value and growth: they do not spread the characteristics that matter
- ▣ Failing to reject is weak evidence; evaluate models by how well they price assets with dispersion in expected returns

Arbitrage Pricing Theory

- ▣ **APT**: Arbitrage Pricing Theory, [Ross \(1976\)](#)
- ▣ Returns follow a K -factor model with small idiosyncratic risks

$$R_{i,t} = \mathbb{E}[R_i] + \beta_i^\top f_t + \varepsilon_{i,t}, \quad \mathbb{E}[f_t] = 0$$

- ▣ **No arbitrage** (no riskless profit without investment) implies, approximately

$$\mathbb{E}[R_i] - R_f = \beta_i^\top \lambda$$

- ▣ λ : the **risk premia**, the extra expected return per unit of each exposure
- ▣ The theory does not say **which** factors: they must be found in the data

Building Factors: Long–Short Portfolios

size: NYSE median; B/M: 30% / 40% / 30%

	Low (growth)	Neutral	High (value)
Small	S_L	S_N	S_H
Big	B_L	B_N	B_H

$$\text{SMB} = \frac{1}{3}(S_L + S_N + S_H) - \frac{1}{3}(B_L + B_N + B_H)$$

$$\text{HML} = \frac{1}{2}(S_H + B_H) - \frac{1}{2}(S_L + B_L)$$

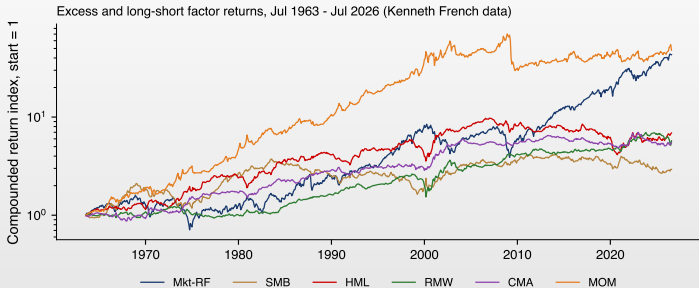
- A **characteristic-sorted factor** buys stocks with a high value of a characteristic and sells those with a low value
- Zero net investment: the return is a pure spread,
 $f_t = R_{\text{high},t} - R_{\text{low},t}$
- Fama–French 2×3 sort each June, accounting data of the previous year, value-weighted portfolios
 - ▶ **SMB** (Small Minus Big): small minus big stocks
 - ▶ **HML** (High Minus Low): high B/M (value) minus low B/M (growth)
- The same construction on profitability and investment gives RMW and CMA (next slide)

The Standard Factor Models

Model	Factors	Reference	Motivation
CAPM	Mkt-RF	Sharpe (1964)	equilibrium
FF3	Mkt-RF, SMB, HML	Fama & French (1993)	size and B/M effects (Fama & French (1992))
Carhart q-factor	FF3 + MOM market, size, investment, ROE	Carhart (1997) Hou, Xue & Zhang (2015)	fund evaluation firms' investment decisions
FF5	Mkt-RF, SMB, HML, RMW, CMA	Fama & French (2015)	dividend-discount model

- FF3: $R_{i,t}^e = \alpha_i + b_i \text{MKT}_t + s_i \text{SMB}_t + h_i \text{HML}_t + \varepsilon_{i,t}$; a good model leaves time-series alphas close to zero (GRS not rejected)
- **RMW** (Robust Minus Weak): high minus low operating profitability; **CMA** (Conservative Minus Aggressive): low minus high asset growth; ROE: Return On Equity
- **MOM** (UMD, Up Minus Down): high minus low return from month $t - 12$ to $t - 2$; past winners keep winning over 3–12 months (Jegadeesh & Titman (1993))
- FF3 uses its own SMB (2×3 sorts on size and B/M); the FF5 SMB averages size spreads from three sorts, so FF3 is not nested in FF5

Six Factors over Sixty-Three Years

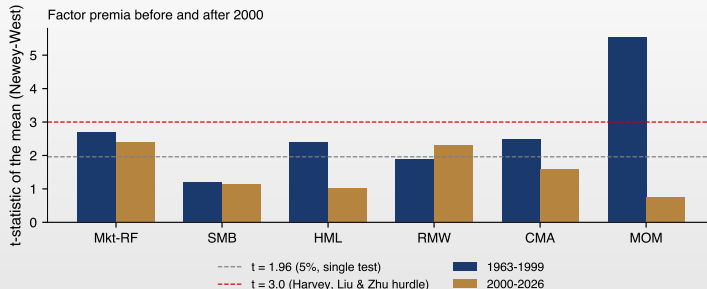


- Compounded monthly factor returns from 1, log scale: excess or long-short returns, so no interest on capital or collateral; not the wealth of a funded investment
- July 1963 – July 2026: Mkt-RF 43.2, MOM 48.0, HML 6.9, RMW 5.7, CMA 5.5, SMB 2.9

- Momentum peaked before 2009 and lost heavily in the 2009 rebound: a **momentum crash**
- Value was flat or losing from about 2007 to 2020
- Average annual returns: Mkt-RF 7.2%, MOM 7.3%, HML 3.6%, RMW 3.1%, CMA 3.0%, SMB 2.3%

 MFM_ch3_factor_cum

Do the Premia Survive?



- t -statistic of the mean monthly factor return (Newey–West), 1963–1999 vs 2000–2026
- Momentum: $t = 5.53$ before 2000, 0.75 after; mean 10.7% vs 2.5% a year
- Value (HML): $t = 2.39$ then 1.02; mean 4.2% vs 2.8%
- Profitability (RMW) is the exception: $t = 1.90$ then 2.31; the market premium holds in both halves ($t = 2.70$ and 2.39)
- A calendar split is not a publication event: dating the decay needs the design of Section 12

The Two Passes



Eugene Fama, Morgan Stanley–AFA

Award, 2008

Photo: TonyTheTiger (2008); CC

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- ▣ Fama & MacBeth (1973) estimate the risk premia λ of $\mathbb{E}[R_i^e] = \lambda_0 + \beta_i^\top \lambda$ (Fama: Nobel Prize 2013)
- ▣ **Pass 1** (time series, one regression per asset): returns $T \times N$, factors $T \times K \Rightarrow$ betas $\hat{B}$, $N \times K$

$$R_{i,t}^e = a_i + \beta_i^\top f_t + \varepsilon_{i,t}$$

- ▣ **Pass 2** (cross section, one regression per month): N returns on $X = [1, \hat{B}]$

$$R_{i,t}^e = \lambda_{0,t} + \hat{\beta}_i^\top \lambda_t + u_{i,t}, \quad \hat{\lambda} = \frac{1}{T} \sum_t \hat{\lambda}_t$$

- ▣ λ_0 : pricing error common to all assets, zero if the model is right
- ▣ Allows factors that are **not** traded portfolios, and firm characteristics as regressors

Standard Errors for Risk Premia

- ▣ **Fama–MacBeth**: treat the monthly $\hat{\lambda}_{k,t}$ as a sample of size T , $se_{FM}(\hat{\lambda}_k) = s(\hat{\lambda}_{k,t})/\sqrt{T}$; robust to cross-sectional correlation of residuals; if $\hat{\lambda}_{k,t}$ is autocorrelated, use Newey–West on this series
- ▣ **Errors in variables**: pass 2 uses $\hat{\beta}$, not β , so se_{FM} is too small; [Shanken \(1992\)](#) scales the non-factor part by $(1 + c)$

$$\text{Var}_{Sh}(\hat{\lambda}) = (1 + c) \left[\text{Var}_{FM}(\hat{\lambda}) - \frac{1}{T} \Sigma_f \right] + \frac{1}{T} \Sigma_f, \quad c = \hat{\lambda}^\top \Sigma_f^{-1} \hat{\lambda}$$

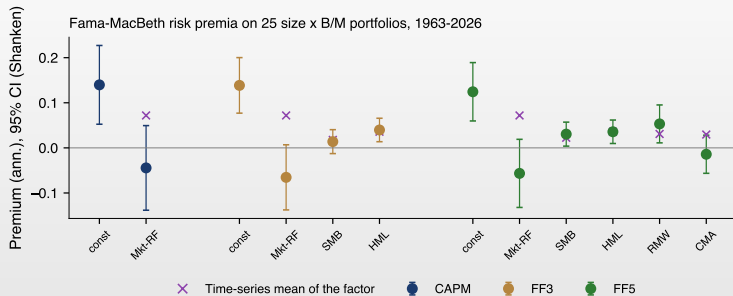
- ▶ Σ_f : factor covariance; c : squared Sharpe ratio of the premia per period, 0.03 for FF3 here, larger for some non-traded factors
- ▣ **Misspecification**: all three assume a correct model, pricing errors $e = \mu - X\gamma = 0$; when GRS rejects, the noise in $\hat{B}$ multiplies e and adds variance ([Kan, Robotti & Shanken \(2013\)](#); GMM form in the [Appendix](#))

Worked Example: Four Standard Errors for the Value Premium

FF3, HML	$\hat{\lambda}$	s.e.	t	Significant at 5%?
Fama–MacBeth	3.97%	1.33%	2.99	yes
Newey–West (6 lags)	3.97%	1.62%	2.45	yes
Shanken	3.97%	1.33%	2.99	yes
Kan–Robotti–Shanken	3.97%	1.32%	3.00	yes

- 25 size $\times$ B/M portfolios, 1963–2026; Shanken barely moves the error ($c = 0.03$ per month)
- Autocorrelation of the monthly slopes $\hat{\lambda}_t$ matters most here: Newey–West, 6 lags by the rule for $T = 757$
- FF5: the misspecification-robust s.e. of RMW rises from 2.1% to 2.6%, and its t falls from 2.5 to 2.1
- At the $t > 3$ hurdle of Section 8, HML is borderline under every choice (2.45–3.00)

Estimated Risk Premia



- 25 size $\times$ B/M portfolios, 1963–2026; bars: 95% intervals with Shanken errors; crosses: factor means
- CAPM: intercept 14.0% (se 4.4%), market premium –4.4% (se 4.8%)
- FF3: SMB 1.4% (1.4%), HML 4.0% (1.3%); FF5: RMW 5.3% (2.1%), CMA –1.4% (2.2%)

- For a traded factor, λ should equal its time-series mean: HML 4.0% vs 3.6%, close; market (FF3) –6.5% vs 7.2%, far
- Market betas span only 0.86–1.42 (cross-sectional s.d. 0.14): λ_m is **weakly identified**, and a large intercept absorbs the level (Lewellen, Nagel & Shanken (2010))

Useless and Weak Factors

- A **useless factor** g_t is independent of all returns: $\beta_g = 0$, so $\hat{\beta}_{i,g} = O_p(T^{-1/2})$ is pure noise
- Pass 2 divides by this noise: $\hat{\lambda}_g$ does not converge to zero and its t -test over-rejects when the model is misspecified ([Kan & Zhang \(1999\)](#))
- Simulation on the 25 portfolios, CAPM plus an independent Normal factor, 2000 replications, nominal level 5%
 - ▶ $\lambda_g = 0$ rejected in 60% (Fama–MacBeth errors) and 27% (Shanken errors) of the replications
 - ▶ First-pass Wald test of $\beta_{i,g} = 0$ for all i : rejected in 6.7%; run it before pass 2
- **Weak factors** (little dispersion in betas, like the market here): identification-robust tests ([Kleibergen \(2009\)](#)); with many assets, consistency of pass 2 depends on factor strength ([Anatolyev & Mikusheva \(2022\)](#))

A Skeptical Appraisal: Adding 30 Industry Portfolios

Model	OLS R^2		GLS R^2	
	25	25 + 30	25	25 + 30
CAPM	0.08	0.00	0.06	0.01
FF3	0.66	0.22	0.21	0.08
FF5	0.76	0.24	0.26	0.10

Monthly 1963–2026, free intercept and premia

- The 25 portfolios have a strong three-factor structure: any factors correlated with SMB and HML fit their means, so a high OLS R^2 is a low hurdle ([Lewellen, Nagel & Shanken \(2010\)](#))
- Three checks
 - ▶ Broaden the test assets: FF3 falls from 0.66 to 0.22 once 30 industries are added
 - ▶ Report the **GLS** (Generalised Least Squares) R^2 , weighted by $\hat{V}^{-1}$ like GRS: never above 0.26
 - ▶ Impose $\lambda = \mathbb{E}[f]$ for traded factors, no intercept: FF3 on the 55 assets gives $R^2 = -0.27$
- Stocks instead of portfolios: more dispersion in betas, larger errors in variables ([Ang, Liu & Schwarz \(2020\)](#))

Factors as a Stochastic Discount Factor



Lars Peter Hansen, Nobel Prize in
Economics 2013

Photo: Bengt Nyman (2013); CC BY
2.0; Wikimedia Commons

- **SDF** (Stochastic Discount Factor) m_t : prices all excess returns, $\mathbb{E}[m_t R_{i,t}^e] = 0$
- A linear factor model is an SDF, $m_t = 1 - \mathbf{b}^\top (\mathbf{f}_t - \mathbb{E}\mathbf{f})$
- Substitute: $\mathbb{E}[R_i^e] = \text{Cov}(R_i^e, \mathbf{f}_t)^\top \mathbf{b} = \beta_i^\top \boldsymbol{\lambda}$ with $\boldsymbol{\lambda} = \boldsymbol{\Sigma}_f \mathbf{b}$: the beta pricing model of Section 6
- $\mathbf{b} \neq 0$ asks whether a factor helps price the assets; $\boldsymbol{\lambda} \neq 0$ whether it is rewarded
- These moment conditions are estimated by GMM, [Hansen \(1982\)](#); survey of the discount-rate view: [Cochrane \(2011\)](#)

Comparing Models: Maximum Sharpe Ratios

Max. Sharpe ratio (annual)	CAPM	FF3	Carhart	FF5	FF5 + MOM
1963–2026 ($T = 757$)	0.47	0.66	0.97	0.97	1.16
2000–2026 ($T = 319$)	0.48	0.54	0.67	0.97	1.02

- Barillas & Shanken (2018): judge a model by $SR^2(f) = \mu_f^\top \Sigma_f^{-1} \mu_f$ of its factors; the test assets do not matter for the comparison
- Nested models: in-sample SR^2 can only rise, so test the alpha of the added factor on the others (spanning regression)
 - HML on the other five factors: $\alpha = 0.9\%$ a year ($t = 0.75$); since 2000 -1.7% ($t = -0.81$): no significant incremental contribution of HML, as in Fama & French (2015)
- Non-nested FF5 vs Carhart: difference in annual SR^2 0.00, 95% interval $[-0.57, 0.66]$; since 2000 0.50, $[-0.24, 1.48]$ (stationary bootstrap, Politis & Romano (1994))
- Another model-independent metric, the Hansen–Jagannathan distance: ► Appendix

The Factor Zoo

- ▣ Cochrane (2011) called the growing list of new factors a “zoo”
- ▣ Harvey, Liu & Zhu (2016) catalogue hundreds of published factors (316 in their list)
- ▣ Each was tested as if it were the only one: $|t| > 1.96$ at 5%
- ▣ With many tests, some pass by chance: **data mining**, also called p-hacking
- ▣ Their proposal: a new factor needs $|t| > 3.0$

Controlling False Discoveries

- **FDR** (False Discovery Rate): expected share of false rejections among all rejections; **FWER** (Family-Wise Error Rate): probability of at least one false rejection ([Chapter 2](#))
- **BH procedure** ([Benjamini & Hochberg \(1995\)](#)): find the largest j with $p_{(j)} \leq \frac{j}{M} \alpha$ and reject $H_{(1)}, \dots, H_{(j)}$

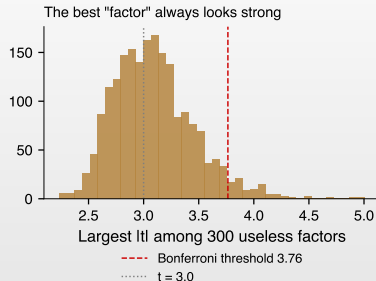
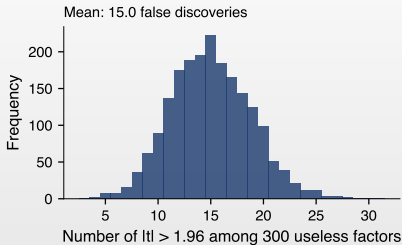
Procedure	Controls	Reject $H_{(j)}$ if	Valid under
Bonferroni	FWER	$p_{(j)} \leq \alpha/M$	any dependence
Holm (Holm (1979))	FWER	$p_{(k)} \leq \alpha/(M - k + 1)$ for all $k \leq j$	any dependence
BH (Benjamini & Hochberg (1995))	FDR	$j \leq \text{largest } k \text{ with } p_{(k)} \leq k\alpha/M$	independence, positive dep.
BY (Benjamini & Yekutieli (2001))	FDR	as BH, thresholds divided by $\sum_{k=1}^M 1/k$	any dependence

- Positive dependence: the test statistics of the true nulls move up together (the Benjamini–Yekutieli PRDS condition: positive regression dependence on a subset); FDR control suits a search over many candidate factors

Worked Example: Ten Candidate Factors

- Sorted p -values: 0.0004, 0.002, 0.006, 0.011, 0.018, 0.027, 0.041, 0.070, 0.15, 0.62; $\alpha = 5\%$
- Naive ($p \leq 0.05$): **7** discoveries
- Bonferroni ($p \leq 0.005$): **2**
- Holm: $0.0004 \leq 0.005$, $0.002 \leq 0.0056$, $0.006 \leq 0.00625$, $0.011 > 0.0071$ stop: **3**
- BH thresholds $0.005j$: $p_{(6)} = 0.027 \leq 0.030$ is the last success: **6**
- Seminar 3, problem A5 repeats the procedure on a different vector, with a threshold chart

Simulation: 300 Useless Factors



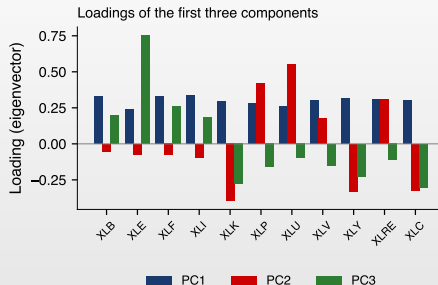
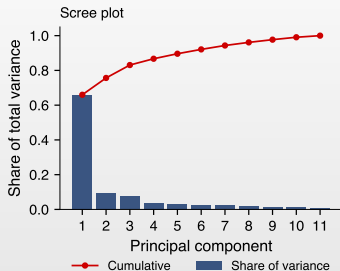
- 2000 replications; each draws 300 independent t -statistics from the standard Normal distribution (no true premium)
- On average 15.0 useless factors pass $|t| > 1.96$; expected value $300 \times 0.05 = 15$

- On average 0.83 pass $|t| > 3$; at least one passes in 57% of the replications; median largest $|t|$: 3.06
- Bonferroni threshold for 300 tests: $|t| > 3.76$; hurdles rise with the number of tests actually tried, including unpublished ones

Principal Components in Brief

- ▣ **Latent factors**: estimated from returns alone, without naming them in advance
- ▣ **PCA** (Principal Component Analysis) of the correlation matrix of standardised returns: $C = \Gamma \Lambda \Gamma^\top$, components $y_t = \Gamma^\top z_t$
- ▣ Eigenvalues $\lambda_1 \geq \dots \geq \lambda_N$; eigenvectors are the **loadings**; component k explains λ_k/N of the variance
- ▣ Two assets with correlation ρ : $\lambda = 1 \pm \rho$, eigenvectors $(1, 1)/\sqrt{2}$ (the “market”) and $(1, -1)/\sqrt{2}$ (the spread); in a crisis ρ rises and PC1 dominates
- ▣ Textbook: [Härdle & Simar \(2019\)](#), chapter on principal components; N equicorrelated assets: Seminar 3, problem A8

PCA of the US Sector ETFs



- 11 sector ETFs, daily log returns, 20 June 2018 – 18 September 2026 ($T = 2073$ common days)
- PC1 explains 66.0%; all loadings positive (0.24–0.34); correlation with SPY 0.954: strongly market-related

- PC2 (9.7%): utilities +0.55, staples +0.42, real estate +0.31 vs technology -0.40, consumer discretionary -0.33: **defensive versus growth**, with no name given in advance
- PC3 (7.4%) is dominated by energy; three components explain 83.1%

How Many Factors?

- Approximate factor model $x_{it} = \lambda_i^\top F_t + e_{it}$, $N, T \rightarrow \infty$; $V(k)$: mean squared residual after the first k components
- Bai & Ng (2002): $\hat{k} = \arg \min_k \left[\ln V(k) + k \frac{N+T}{NT} \ln \min(N, T) \right]$ (criterion IC_{p2}); Ahn & Horenstein (2013): eigenvalue ratio $\hat{k} = \arg \max_k \mu_k / \mu_{k+1}$, no penalty to choose
- Kaiser ($\lambda_k > 1$) has no asymptotic justification: the noise eigenvalues of a correlation matrix scatter around 1

Panel	N	T	Kaiser	Bai–Ng	Ahn–Horenstein
11 sector ETFs, daily 2018–2026	11	2073	2	5	1
100 size $\times$ B/M portfolios, monthly 1963–2026	95	757	3	4	1

- Standardised returns; maximum k : 5 for sectors, 8 for portfolios; 5 portfolios with missing months are dropped
- Ahn–Horenstein picks the dominant market factor (ratio 14.6); its next peak is at $k = 3$ (ratio 3.5), the FF3 structure
- Bai–Ng on 11 sectors hits the upper bound $k = 5$: the penalty is built for large N , so with small N the answer is fragile

Limits of PCA for Asset Pricing

- ▣ Loadings are **constant**: PCA cannot follow a stock whose size or value changes
- ▣ Components explain **variance**, not necessarily **average returns**
- ▣ Individual stocks enter and exit the sample: a stock panel is unbalanced
- ▣ Component signs are arbitrary; rotations preserve principal components only within repeated-eigenvalue eigenspaces, while a latent-factor model is identified only up to rotation
- ▣ Risk-premium PCA and three-pass premia keep the PCA backbone but target pricing (▶ [Appendix](#)); PCA and factor analysis in finance: [Franke, Härdle & Hafner \(2019\)](#)

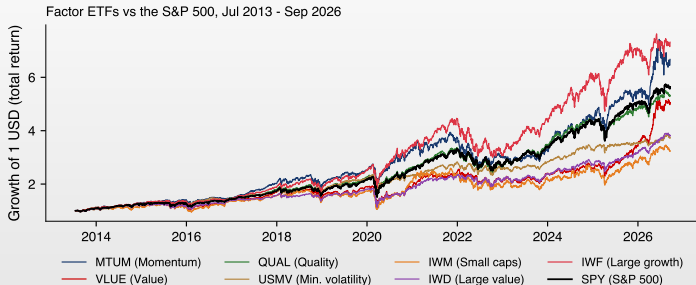
Instrumented PCA

- ▣ **IPCA:** Instrumented Principal Component Analysis, [Kelly, Pruitt & Su \(2019\)](#)
- ▣ Loadings are functions of L observed characteristics $\mathbf{z}_{i,t}$ (size, B/M, momentum, ...)

$$R_{i,t+1}^e = \alpha_{i,t} + \boldsymbol{\beta}_{i,t}^\top \mathbf{f}_{t+1} + \varepsilon_{i,t+1}, \quad \boldsymbol{\beta}_{i,t}^\top = \mathbf{z}_{i,t}^\top \boldsymbol{\Gamma}_\beta$$

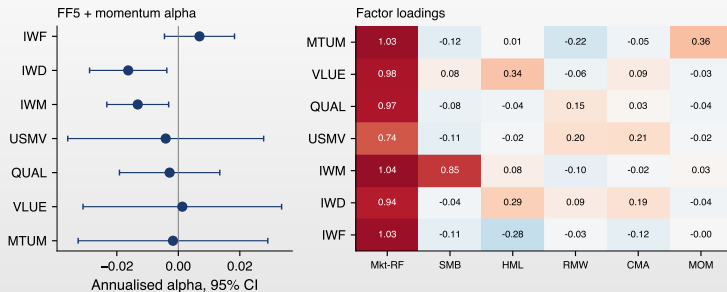
- ▣ $\boldsymbol{\Gamma}_\beta$ ($L \times K$) and the latent factors $\mathbf{f}_t$ are estimated jointly by alternating least squares
- ▣ Test: are characteristics priced through betas ("covariances"), or directly as alphas ($\alpha_{i,t} = \mathbf{z}_{i,t}^\top \boldsymbol{\Gamma}_\alpha$)?
- ▣ Result in their US stock data: a few IPCA factors price the cross section better than observable-factor models with as many factors, and $\boldsymbol{\Gamma}_\alpha$ is insignificant: "characteristics are covariances"

Factor ETFs versus the Market



- **Smart beta:** rule-based long-only portfolios that tilt towards factors, sold as ETFs; they keep market exposure and need not replicate an academic long-short factor
- Growth of 1 USD with dividends re-invested, from 18 July 2013 (first common day)
- **CAGR:** large growth (IWF) 16.3% and momentum (MTUM) 15.5% beat SPY (14.0%); quality (QUAL) 13.5%, value (VLUE) 13.1%, large value (IWD) 10.7%, minimum volatility (USMV) 10.5%, small caps (IWM, Russell 2000 index) 9.3%
- Thirteen years of one realisation: not evidence against the premia on its own

Alphas of Factor ETFs



- Daily regressions on FF5 + MOM, Newey–West errors, each ETF on its own trading calendar (2000–2026)
- The loadings match the labels: MTUM on MOM 0.36, VLUE on HML 0.34, IWM on SMB 0.85; IWF -0.28 on HML; USMV market beta 0.74

- Alphas close to zero: MTUM -0.2% ($t = -0.11$), QUAL -0.3%; Russell ETFs IWM -1.3% ($t = -2.57$), IWD -1.6% ($t = -2.55$), consistent with costs and index construction
- R^2 between 0.81 and 0.97: mostly repackaged factor exposure

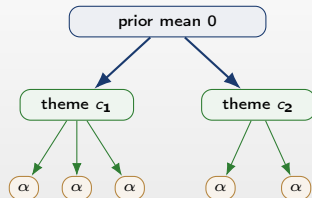
Timing, Crowding and Costs

- **Factor timing:** switching between factors based on signals (valuation spreads, recent returns)
 - ▶ Noisy premia and short histories make timing hard to verify out of sample ([Chapter 8](#))
- **Crowding:** many investors hold the same factor positions
 - ▶ Premia can shrink ([McLean & Pontiff \(2016\)](#)), and crowded positions can amplify losses during forced unwinding; the 2009 momentum crash coincided with a sharp rebound of past losers ([Daniel & Moskowitz \(2016\)](#))
- **Implementability:** long–short factors ignore trading costs and short-selling fees; ETFs pay them

Case Study: Research Question

- **The challenge** to factor research takes two forms (Jensen, Kelly & Pedersen (2023), Introduction)
 - ▶ No internal validity: results cannot be reproduced; Hou, Xue & Zhang (2020) replicate only 35% of their anomalies
 - ▶ No external validity: results come from testing many factors without controlling the FDR; Harvey, Liu & Zhu (2016) conclude that most claimed findings are likely false
- **Question:** do published factors have significant CAPM alphas in the original data, out of sample and in other countries, once multiple testing is taken into account?
- **Why it matters:** if most factors are false discoveries, multifactor models and factor investing (Sections 5 and 10) rest on noise
- **A second view of the zoo:** most new factors are redundant given the existing ones (Feng, Giglio & Xiu (2020),  Appendix); the paper instead treats the zoo as correlated evidence on a few economic themes
- **Link to this chapter:** Jensen's alpha (Section 4) is the object tested; Benjamini–Yekutieli (Section 8) is the frequentist benchmark; Bayesian shrinkage extends the Vasicek beta (Section 3) to alphas

Case Study: Data and Method



each factor's alpha borrows strength from the other factors of its theme

- **Data** (Section II): 153 factors in 93 countries; terciles of each characteristic, capped value weights, long the side with the higher expected return in the original paper; each factor scaled to a 10% annual idiosyncratic volatility
- **Themes**: 13 clusters of US factors, hierarchical clustering on the correlation of CAPM residuals
- **Model** (Section I): $f_{i,t} = \alpha_i + \beta_i r_t^m + \varepsilon_{i,t}$, alphas $\alpha_i = c_j + s_n + w_i$ (theme, signal, factor), prior mean 0
- **One factor**: $\mathbb{E}(\alpha \mid \hat{\alpha}) = \kappa \hat{\alpha}$, $\kappa = 1/(1 + \sigma^2/(\tau^2 T))$ (eq. 5): short, noisy samples are shrunk more, as in Vasicek
- **Replication rate**: share of factors with a significant alpha; Bayesian discovery if posterior mean / posterior s.d. ≥ 1.96
- **Full hierarchy and estimation**: [▶ Appendix](#)

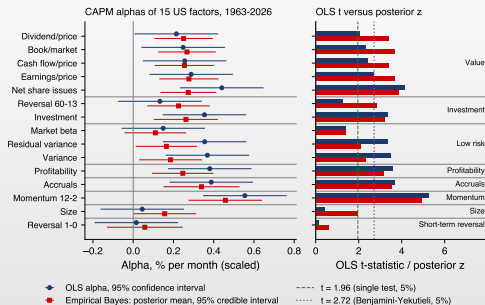
Case Study: Key Results

Replication rates, US (Figure 1)

Hou, Xue & Zhang (2020), raw returns	35.0%
This paper, raw returns	55.6%
CAPM alpha, OLS	82.4%
CAPM alpha, Benjamini–Yekutieli	75.6%
CAPM alpha, empirical Bayes	82.4%
World, empirical Bayes	82.4%

- Testing alphas instead of raw returns drives most of the gap with Hou, Xue & Zhang (2020)
- Bayesian FDR 0.1%: shrinkage controls false discoveries without the power loss of a higher t hurdle
- **Decay**: mean alpha 0.49% a month in sample, 0.26% after the original sample (−47%); slope of post-sample on in-sample alphas 0.26 (Figure 8), against about 0.9 from shrinkage alone
- The strongest factors decay more: arbitrage or data mining (Section 12)
- Complete results: [▶ Appendix](#)

Case Study: Replication on Course Data



- 15 US long-short factors from value-weighted sorts of the [Kenneth French Data Library](#), July 1963 – July 2026 ($T = 757$); CAPM alphas scaled to 10% idiosyncratic volatility; themes of the paper ([authors' cluster labels](#))
- Replication rates: OLS 73.3% (11 of 15), Benjamini-Yekutieli 53.3% (8; critical $t = 2.72$), empirical Bayes 86.7% (13)

- Shrinkage towards the theme, not only towards zero: long-term reversal (investment theme) $0.13\% \rightarrow 0.23\%$ a month ($t = 1.25$, posterior $z = 2.85$)
- Failures: market beta and short-term reversal ($z \leq 1.41$); size only just passes ($z = 1.96$), echoing the paper, where size is one of the two themes below 50%

Case Study: Limits and an Open Question

▣ Limits of the course version

- ▶ 15 US signals instead of 153 in 93 countries; Kenneth French Data Library sort breakpoints instead of non-micro terciles with capped weights; one region, so no signal level ($\tau_s = 0$)
- ▶ The rate depends on the theme structure: with a single common theme the empirical-Bayes rate is 100%

▣ Limits stated by the authors: CAPM as the only benchmark, uniform construction of the characteristics, unobserved factors in the prior (▶ Appendix)

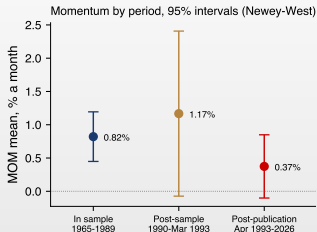
▣ Open question: the paper's data end in 2020; did the posterior alphas and the replication rate hold in 2021–2026, and can a prior estimated on US factors sharpen short-sample BVB factors?

▣ Data and code of the paper: jkpfactors.com; course project: Seminar 3, C3

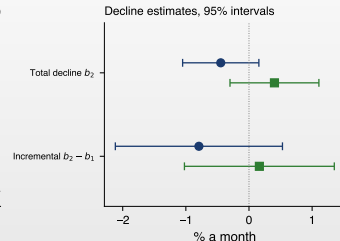
An Open Question and the Discovery Loop

- **Open question:** why do factor premia decay after publication: arbitrage, data mining or a change in risk, and does it differ across factors and markets?
- Evidence so far
 - ▶ [McLean & Pontiff \(2016\)](#): across 97 predictors, returns are 26% lower out of sample and 58% lower after publication; the gap is attributed to publication-informed trading
 - ▶ [Jensen, Kelly & Pedersen \(2023\)](#): most factors replicate, but the strongest decay most; [Chen & Zimmermann \(2022\)](#): open data and code for the published predictors
- **One pass of the loop**
 - ▶ The assistant drafts: a literature table with DOIs, competing hypotheses with their predicted patterns, the regression code, a robustness plan
 - ▶ The human fixes before running: the event dates (Jegadeesh–Titman: sample ends 1989, published March 1993), the factor set, the multiple-testing correction, the placebo
 - ▶ Checks: every DOI resolves; the chapter's split-sample t -statistics are reproduced first; the data vintage is recorded
- Reusable prompts and the full checklist: [▶ Appendix](#)

Mini-Case: Was Momentum Arbitraged Away?



MOM



Mkt-RF (placebo)

- Design of McLean & Pontiff (2016):
 $MOM_t = a + b_1 \text{PostSample}_t + b_2 \text{PostPub}_t + e_t$, monthly 1965–2026, Newey–West errors; post-sample 1990-01 – 1993-03, post-publication from 1993-04
- b_2 : total decline relative to the in-sample mean; $b_2 - b_1$: incremental effect of publication

- $a = 0.82\%$ a month ($t = 4.32$); $b_2 = -0.45\%$ ($t = -1.45$, interval $[-1.05\%, 0.16\%]$): 54% lower; $b_2 - b_1 = -0.79\%$ ($t = -1.18$); placebo Mkt-RF $b_2 = +0.40\%$ ($t = 1.12$)
- Close to the 58% of McLean & Pontiff (2016), but one factor cannot reject $H_0 : b_2 = 0$ at 5%; arbitrage is one interpretation, not an identified cause

Your Turn: A Project Seed

- **Question:** do the Fama–French factors and momentum decay after publication, and does the decay differ across factors?
 - ▶ Pooled regression of SMB, HML, RMW, CMA and MOM on factor-specific post-sample and post-publication dummies; Mkt-RF as placebo; errors robust to cross-factor correlation and autocorrelation
 - ▶ Whether decay is larger where arbitrage is easier needs stock-level proxies (size, liquidity, short-sale costs), not factor returns
- **Replicate first:** the split-sample t -statistics and the mini-case estimates
- **Deliverables:** a pre-registration note (H_0 , H_1 , dates from each paper, factor set, corrections) committed before the results; code as a Quantlet; a report with Newey–West inference, placebo periods and power
- **Logs:** prompts in `AI_USE.md`; every caught error (a wrong publication date, a misquoted number) in `AI_ERRORS.md`

Key Takeaways: An Empirical Workflow

1. **Estimate exposures** with HAC errors, report sample and proxy, shrink extreme betas (Blume, Vasicek)
 2. **Test the pricing restriction jointly**: GRS measures the Sharpe-ratio gain from the test assets; the CAPM fails on the 25 size $\times$ B/M portfolios (flat SML, $GRS = 4.20$)
 3. **Estimate premia** in two passes with errors robust to estimated betas and misspecification; a high R^2 on 25 portfolios is a low hurdle
 4. **Guard against the zoo**: multiple-testing hurdles, spanning tests, model comparison by maximum Sharpe ratios
 5. **Replicate**: latent factors and out-of-sample evidence; factor ETFs are cheap factor exposure, not alpha
- Open: why premia decay after publication; how to price with time-varying betas; which factors price assets in small markets such as the BVB

Check Yourself, and Next

Retrieval questions

- Why does the HAC standard error of a beta differ from the classical one, while the beta does not?
- Why can GRS fail to reject the CAPM on sector ETFs but reject it on the 25 portfolios over the same months?
- Why does a useless factor look priced in pass 2?
- Why can the empirical-Bayes replication rate exceed the Benjamini–Yekutieli rate?

Links to other chapters

- [Chapter 2](#): an alpha is an anomaly or a missing risk factor, the joint hypothesis problem
- [Chapter 4](#): factor covariance models and shrinkage for Markowitz
- [Chapters 5–6](#): time-varying volatility, correlation and betas
- [Chapter 8](#): backtesting factor strategies without overfitting
- [Chapter 13](#): machine learning for the cross section of returns

Appendix: GRS under the Null Hypothesis

- Under H_0 and the exact assumptions, $W \sim F(N, T - N - 1)$, so $\mathbb{E}[W] = \frac{T-N-1}{T-N-3}$
- Invert the Sharpe-ratio form of W : the squared Sharpe-ratio gain expected under H_0 is

$$\mathbb{E}[\widehat{SR}^2(f, R) - \widehat{SR}^2(f)] \approx \frac{N(1 + \widehat{SR}^2(f))}{T - N - 3}$$

- 25 portfolios, $T = 757$: $25 \times 1.018/729 = 0.035$ (monthly), against $0.405^2 - 0.134^2 = 0.146$ observed
- In-sample maximisation over N assets always adds some Sharpe ratio: the test asks whether the gain exceeds this noise level
- 9 sector ETFs, $T = 331$: observed gain $0.191^2 - 0.146^2 = 0.015$, below the expected $9 \times 1.021/319 = 0.029$

Appendix: Tests with Many Assets

- $\hat{\Sigma} = \hat{E}^\top \hat{E} / T$ has rank at most $T - 2$ (two parameters per regression)
 - ▶ Example: 100 size $\times$ B/M portfolios on 8 years of monthly data ($T = 96$): $\hat{\Sigma}$ is singular, GRS does not exist
 - ▶ Well before this point the inverse is badly estimated and power falls
- Tests for many assets that avoid $\hat{\Sigma}^{-1}$: standardised sums of squared t -statistics, [Pesaran & Yamagata \(2024\)](#)
- Power enhancement by screening large alphas: [Fan, Liao & Yao \(2015\)](#)
- With many assets, consistency of the second pass depends on factor strength: [Anatolyev & Mikusheva \(2022\)](#)

Appendix: Misspecification-Robust Standard Errors

- Fama–MacBeth and Shanken errors assume a **correct** model: all pricing errors $e = \mu - X\gamma$ are zero ($X = [1, B]$)
- GRS rejects here, so γ is a pseudo-true value; the noise in $\hat{B}$ multiplies e and adds a variance term ([Kan, Robotti & Shanken \(2013\)](#))
- Write both passes as one exactly identified GMM system: moments $(R_t - a - Bf_t) \otimes (1, f_t)$ and $X^\top (R_t - X\gamma)$; the sandwich variance with the full Jacobian is the robust one
 - ▶ Setting $e = 0$ in the Jacobian gives the correct-model, heteroskedasticity-robust version of [Jagannathan & Wang \(1998\)](#)
- FF5 on the 25 portfolios: s.e. of RMW 2.1% $\rightarrow$ 2.6%, CMA 2.2% $\rightarrow$ 2.7%, intercept 3.3% $\rightarrow$ 3.8%
- RMW: t falls from 2.5 to 2.1; HML changes by less than 0.05 percentage points
- Advanced course:  [Generalized Method of Moments](#)

Appendix: Hansen–Jagannathan Distance and Bayesian Selection

- **Hansen–Jagannathan distance** (Hansen & Jagannathan (1997)): least-squares distance of the model SDF from the set of valid SDFs

$$\delta^2 = \min_b e(b)^\top \mathbb{E}[R^e R^{e\top}]^{-1} e(b), \quad e(b) = \mathbb{E}[m_t(b) R_t^e]$$

- ▶ Excess returns R_t^e with the normalisation $\mathbb{E}[m] = 1$; for gross returns R_t : $e = \mathbb{E}[mR] - 1$, and the SDF needs a free scale ($\mathbb{E}[m] = 1/(1 + R_f)$)
- ▶ Same weighting matrix for every model, so distances are comparable; the test statistic has a weighted χ^2 limit
- Zoo-scale selection: Bayesian averaging over two quadrillion factor models, robust to weak factors (Bryzgalova, Huang & Julliard (2023))
- SDF shrinkage on characteristic-based factors: a few principal components of anomaly returns capture the SDF (Kozak, Nagel & Santosh (2020))
- Common lesson: compare models by a single, model-independent metric, with its sampling uncertainty

Appendix: Taming the Zoo with Double-Selection LASSO

- Feng, Giglio & Xiu (2020): does a new factor add information given the existing zoo?
- **Double-selection LASSO** (Least Absolute Shrinkage and Selection Operator)
 - ▶ Step 1: a LASSO of average test-asset returns on their covariances with hundreds of factors selects the controls that price the cross section
 - ▶ Step 2: a second LASSO selects the factors whose covariances explain those of the new factor, so that omitted controls do not bias its SDF coefficient
 - ▶ Step 3: the SDF coefficient of the new factor is estimated with the union of both sets of controls and tested
- Result: most new factors are redundant; only a few add pricing information
- The case study reaches a compatible conclusion: the zoo is highly redundant, and a few themes carry the information

Appendix: More on the Number of Factors

- ▣ Onatski (2010): separates factor eigenvalues from the edge of the noise eigenvalue distribution
- ▣ Kaiser ($\lambda_k > 1$) on a correlation matrix: with N assets and pure noise the eigenvalues spread around 1 with a width that depends on N/T , so the rule has no asymptotic justification
- ▣ Bai–Ng penalties are built for large N and T : with 11 sectors the criterion keeps falling up to $k_{\max}$
- ▣ Ahn–Horenstein needs no penalty: the ratio of consecutive eigenvalues peaks after the last strong factor
- ▣ Report the range of estimates and the sensitivity to $k_{\max}$, not a single number (Seminar 3, problem B3)

Appendix: Risk-Premium PCA and Three-Pass Premia

- ▣ **RP-PCA** (Risk-Premium PCA, [Lettau & Pelger \(2020\)](#)): principal components of $\frac{1}{T}X^T X + \gamma \bar{x}\bar{x}^T$
 - ▶ $\gamma = -1$ gives covariance PCA; $\gamma > -1$ also rewards factors that explain average returns
 - ▶ In their data it finds factors with higher Sharpe ratios and smaller pricing errors than PCA
- ▣ **Three-pass estimator** ([Giglio & Xiu \(2021\)](#)): the premium of an observed factor g_t when the model omits factors
 - ▶ (1) PCA of test-asset returns; (2) cross-sectional regression of means on the latent betas; (3) regress g_t on the latent factors and map their premia to g_t
- ▣ Both keep PCA's statistical backbone but target pricing, not only variance; IPCA adds characteristics

Appendix: IPCA and Crypto-Asset Factors

- Title of the IPCA paper, [Kelly, Pruitt & Su \(2019\)](#): “Characteristics are covariances”
- In their US stock data, a few IPCA factors price the cross section better than observable-factor models with as many factors
- The alpha term Γ_α becomes insignificant: characteristics predict returns because they measure risk exposure
- IPCA links PCA (statistics) with characteristic sorts (Fama–French): time-varying betas, few factors
- Crypto-assets: market, size and momentum factors price the cross section of coins ([Liu, Tsyvinski & Wu \(2022\)](#)); dimension reduction on crypto and traditional return features: [Pele et al. \(2023\)](#), [Jheng et al. \(2026\)](#)
- Advanced course:  [Multivariate Statistical Analysis](#)

Appendix: Case Study, Data and Construction

- **Data** (Section II): 153 factors in 93 countries; 138 from the list of [Hou, Xue & Zhang \(2020\)](#) plus 15 from the wider literature; US returns from 1926, most developed markets from 1986; all in US dollars, in excess of the US Treasury bill
 - ▶ Each country and month: terciles of the characteristic, breakpoints from non-micro stocks; “capped” value weights (market equity winsorised at the NYSE 80th percentile); long the tercile with the higher expected return in the original paper, one-month holding
 - ▶ At least 5 stocks per leg and 60 monthly observations; every factor scaled to a 10% annual idiosyncratic volatility
 - ▶ 13 themes: hierarchical clustering of US factors, distance 1 – correlation of CAPM residuals, Ward linkage (Section II.B)
- **Model** (Section I): $f_{i,t} = \alpha_i + \beta_i r_t^m + \varepsilon_{i,t}$; alphas $\alpha_i = c_j + s_n + w_i$ (theme j , signal n , factor i ; eq. 21), prior mean 0
 - ▶ Prior variance $\Omega = MM^\top \tau_c^2 + ZZ^\top \tau_s^2 + I\tau_w^2$ (eq. 23); posterior mean $(\Omega^{-1} + T\Sigma^{-1})^{-1} T\Sigma^{-1} \hat{\alpha}$ (Proposition 4)
 - ▶ Empirical Bayes: τ_c, τ_s, τ_w by maximum likelihood of $\hat{\alpha} \sim N(0, \Omega + \hat{\Sigma}/T)$ (Section I.C.2)
- **Replication rate**: share of factors significant in the original paper whose alpha is significant here; Bayesian discovery if posterior mean / posterior s.d. ≥ 1.96 (eq. 26)

Appendix: Case Study, Complete Results

- Replication rates (Figure 1): Hou, Xue & Zhang (2020) 35.0%; US raw return 55.6%; originally significant 61.3%; CAPM alpha 82.4%; Benjamini–Yekutieli 75.6%; empirical Bayes 82.4%; world, empirical Bayes 82.4%
- OLS: 98 of 119 US factors; Benjamini–Yekutieli needs $|t| > 2.7$ (Figure 4)
- Bayesian FDR 0.1%, 95% interval $[0, 1\%]$; expected share of true factors 94% (Section III.C)
- Mega and large stocks: 77.3% and 79.8% (Figure 5); 11 of 13 themes at least 50%, all but low leverage and size: not an artefact of micro-caps
- Mean alpha 0.49% a month in sample, 0.26% after the original sample: -47% ; slope of post-sample alphas on in-sample alphas 0.26 (Figure 8), against about 0.9 expected from shrinkage alone
- Tangency Sharpe ratio (monthly), US 1972–2020: market 0.13, all factors 0.80 (Figure 14)

Appendix: Case Study, the Course Replication in Detail

- ▣ Benjamini–Yekutieli is the strictest, as in the US rates of the paper: 82.4%, 75.6%, 82.4% (Figure 1)
- ▣ Net share issues move towards their theme: 0.44% $\rightarrow$ 0.27% a month
- ▣ Posterior s.d. 0.07%–0.10% against OLS s.e. 0.11%: each theme pools the evidence of its members
- ▣ **Prior dispersion:** $\hat{\tau}_c = 0.28\%$, $\hat{\tau}_w = 0.04\%$ a month, against 0.35% and 0.21% in Table I; 15 signals leave little factor-specific dispersion
- ▣ **Themes:** the 15 signals fall into 8 of the paper's 13 themes (5 in value, 3 in low risk, 2 in investment); with a single common theme the empirical-Bayes rate is 100%

Appendix: Case Study, Limits Stated by the Authors and Extensions

▣ Limits stated by the authors

- ▶ CAPM is the only benchmark: low-leverage factors fail on CAPM alpha but matter jointly with other themes (Section III.A)
- ▶ Characteristics are built consistently, sometimes unlike the original paper (Section II.C)
- ▶ Unobserved factors bias the prior: with the scenarios of [Harvey, Liu & Zhu \(2016\)](#) the US rate falls to 81.5% and 79.8% (Figure 10)

▣ Complete the course replication (Seminar 3, C3): justify each sign from the original paper, compare the dendrogram with the paper's themes, add the Bayesian FDR and the post-sample slope (Figure 8)

▣ Data and code of the paper: jkpfactors.com, github.com/bkelly-lab/ReplicationCrisis

Appendix: The Discovery Loop, Reusable Prompts (1/2)

□ Literature

- ▶ Summarise the evidence since 2016 on post-publication decay and replication of anomaly returns. For each paper: DOI, sample, number of factors, headline estimate, identification of the decay. Flag entries you are unsure of.
- ▶ Resolve every DOI on Crossref; check each headline number against the abstract

□ Hypothesis

- ▶ Contrast data mining, publication-informed arbitrage and time-varying risk as explanations of decay. For each, describe the predicted pattern across the in-sample, post-sample and post-publication periods, and across high vs low limits-to-arbitrage stocks.

□ Data and code

- ▶ Load the monthly momentum factor from the Kenneth French Data Library; from 1965, regress MOM on a post-sample dummy (1990-01 to 1993-03) and a post-publication dummy (from 1993-04), with Newey-West errors.
- ▶ Replicate a known number first: momentum $t = 5.53$ before 2000 and 0.75 after (split-sample chart)

Appendix: The Discovery Loop, Reusable Prompts (2/2)

▣ Robustness

- ▶ Write a pre-registered robustness plan: one publication date per factor, a Holm correction across factors, excluding the 2009 momentum crash, the working-paper date instead of the journal date, and 1927-1962 as a placebo period for Mkt-RF, SMB, HML and MOM (RMW and CMA start in July 1963).

▣ Critique

- ▶ Referee this: one factor, one publication date, $t = -1.45$. What is the power to detect a 58% decay, and why does pooling many anomalies change it?

▣ Report

- ▶ In-sample mean, both dummies with Newey–West errors, the placebo, the data vintage (download date)
- ▶ Pooled factors: errors robust to cross-factor correlation within a month and to autocorrelation
- ▶ Prompts that shaped a decision go into AI_USE.md

- ▣ The assistant enumerates specifications; the human fixes which ones count before running them

Appendix: What the Researcher Must Check

- **Wrong event date:** sample end, working paper and journal issue are different dates
 - ▶ test: read them from the paper itself; Jegadeesh–Titman: sample ends 1989, published March 1993
- **Choosing the factor after seeing it decay:** fix the set of factors in advance; Holm or BH across them
- **Serially correlated, crash-prone returns:** Newey–West errors; rerun without 2009
- **Data vintages:** the factor library is revised retroactively
 - ▶ test: record the download date; reproduce the chapter's split-sample t -statistics
- **Misquoted literature:** a common slip gives 58% as the out-of-sample decline; it is post-publication (26% out of sample)
- **Low power:** a single factor rarely rejects; report the confidence interval of the decay, not only its sign

Appendix: Further Study and Code

Advanced courses on quantinar.com

- ▣  Generalized Method of Moments
- ▣  Multivariate Statistical Analysis

Code and data

- ▣ All charts and tables: github.com/danpele/MFM/tree/main/Quantlets/Ch_03
- ▣ Factor data: [Kenneth French Data Library](#)

References (1/7)

- ▣ Ahn, S.C., Horenstein, A.R. (2013). Eigenvalue ratio test for the number of factors. *Econometrica*, 81(3), 1203–1227.
- ▣ Anatolyev, S., Mikusheva, A. (2022). Factor models with many assets: strong factors, weak factors, and the two-pass procedure. *Journal of Econometrics*, 229(1), 103–126.
- ▣ Ang, A., Liu, J., Schwarz, K. (2020). Using stocks or portfolios in tests of factor models. *Journal of Financial and Quantitative Analysis*, 55(3), 709–750.
- ▣ Bai, J., Ng, S. (2002). Determining the number of factors in approximate factor models. *Econometrica*, 70(1), 191–221.
- ▣ Barillas, F., Shanken, J. (2018). Comparing asset pricing models. *Journal of Finance*, 73(2), 715–754.
- ▣ Benjamini, Y., Hochberg, Y. (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society B*, 57(1), 289–300.
- ▣ Black, F. (1972). Capital market equilibrium with restricted borrowing. *Journal of Business*, 45(3), 444–455.
- ▣ Blume, M.E. (1971). On the assessment of risk. *Journal of Finance*, 26(1), 1–10.
- ▣ Benjamini, Y., Yekutieli, D. (2001). The control of the false discovery rate in multiple testing under dependency. *Annals of Statistics*, 29(4), 1165–1188.
- ▣ Bryzgalova, S., Huang, J., Julliard, C. (2023). Bayesian solutions for the factor zoo: we just ran two quadrillion models. *Journal of Finance*, 78(1), 487–557.

References (2/7)

- ▣ Campbell, J.Y., Lo, A.W., MacKinlay, A.C. (1997). *The Econometrics of Financial Markets*. Princeton University Press.
- ▣ Carhart, M.M. (1997). *On persistence in mutual fund performance*. *Journal of Finance*, 52(1), 57–82.
- ▣ Chen, A.Y., Zimmermann, T. (2022). *Open source cross-sectional asset pricing*. *Critical Finance Review*, 11(2), 207–264.
- ▣ Cochrane, J.H. (2011). *Presidential address: discount rates*. *Journal of Finance*, 66(4), 1047–1108.
- ▣ Daniel, K., Moskowitz, T.J. (2016). *Momentum crashes*. *Journal of Financial Economics*, 122(2), 221–247.
- ▣ Fama, E.F., French, K.R. (1992). *The cross-section of expected stock returns*. *Journal of Finance*, 47(2), 427–465.
- ▣ Fama, E.F., French, K.R. (1993). *Common risk factors in the returns on stocks and bonds*. *Journal of Financial Economics*, 33(1), 3–56.
- ▣ Fama, E.F., French, K.R. (2015). *A five-factor asset pricing model*. *Journal of Financial Economics*, 116(1), 1–22.
- ▣ Fama, E.F., MacBeth, J.D. (1973). *Risk, return, and equilibrium: empirical tests*. *Journal of Political Economy*, 81(3), 607–636.
- ▣ Fan, J., Liao, Y., Yao, J. (2015). *Power enhancement in high-dimensional cross-sectional tests*. *Econometrica*, 83(4), 1497–1541.

References (3/7)

- ▣ Feng, G., Giglio, S., Xiu, D. (2020). Taming the factor zoo: a test of new factors. *Journal of Finance*, 75(3), 1327–1370.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- ▣ Frazzini, A., Pedersen, L.H. (2014). Betting against beta. *Journal of Financial Economics*, 111(1), 1–25.
- ▣ Gibbons, M.R., Ross, S.A., Shanken, J. (1989). A test of the efficiency of a given portfolio. *Econometrica*, 57(5), 1121–1152.
- ▣ Giglio, S., Xiu, D. (2021). Asset pricing with omitted factors. *Journal of Political Economy*, 129(7), 1947–1990.
- ▣ Hansen, L.P. (1982). Large sample properties of generalized method of moments estimators. *Econometrica*, 50(4), 1029–1054.
- ▣ Hansen, L.P., Jagannathan, R. (1997). Assessing specification errors in stochastic discount factor models. *Journal of Finance*, 52(2), 557–590.
- ▣ Härdle, W.K., Simar, L. (2019). *Applied Multivariate Statistical Analysis*, 5th ed. Springer.
- ▣ Harvey, C.R., Liu, Y., Zhu, H. (2016). . . . and the cross-section of expected returns. *Review of Financial Studies*, 29(1), 5–68.

References (4/7)

- ▣ Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2), 65–70.
- ▣ Hou, K., Xue, C., Zhang, L. (2015). Digesting anomalies: an investment approach. *Review of Financial Studies*, 28(3), 650–705.
- ▣ Hou, K., Xue, C., Zhang, L. (2020). Replicating anomalies. *Review of Financial Studies*, 33(5), 2019–2133.
- ▣ Jagannathan, R., Wang, Z. (1998). An asymptotic theory for estimating beta-pricing models using cross-sectional regression. *Journal of Finance*, 53(4), 1285–1309.
- ▣ Jegadeesh, N., Titman, S. (1993). Returns to buying winners and selling losers: implications for stock market efficiency. *Journal of Finance*, 48(1), 65–91.
- ▣ Jensen, M.C. (1968). The performance of mutual funds in the period 1945–1964. *Journal of Finance*, 23(2), 389–416.
- ▣ Jensen, T.I., Kelly, B., Pedersen, L.H. (2023). Is there a replication crisis in finance?. *Journal of Finance*, 78(5), 2465–2518.
- ▣ Jheng, S.-L., Găman, Ș., Conda, A., Pele, D.T., Härdle, W.K., Teng, H.-W., Osterrieder, J. (2026). Crypto currency returns. SSRN Working Paper.
- ▣ Kan, R., Robotti, C., Shanken, J. (2013). Pricing model performance and the two-pass cross-sectional regression methodology. *Journal of Finance*, 68(6), 2617–2649.
- ▣ Kan, R., Zhang, C. (1999). Two-pass tests of asset pricing models with useless factors. *Journal of Finance*, 54(1), 203–235.

References (5/7)

- ▣ Kelly, B.T., Pruitt, S., Su, Y. (2019). Characteristics are covariances: a unified model of risk and return. *Journal of Financial Economics*, 134(3), 501–524.
- ▣ Kleibergen, F. (2009). Tests of risk premia in linear factor models. *Journal of Econometrics*, 149(2), 149–173.
- ▣ Kozak, S., Nagel, S., Santosh, S. (2020). Shrinking the cross-section. *Journal of Financial Economics*, 135(2), 271–292.
- ▣ Lettau, M., Pelger, M. (2020). Factors that fit the time series and cross-section of stock returns. *Review of Financial Studies*, 33(5), 2274–2325.
- ▣ Lewellen, J., Nagel, S., Shanken, J. (2010). A skeptical appraisal of asset pricing tests. *Journal of Financial Economics*, 96(2), 175–194.
- ▣ Lintner, J. (1965). The valuation of risk assets and the selection of risky investments in stock portfolios and capital budgets. *Review of Economics and Statistics*, 47(1), 13–37.
- ▣ Liu, Y., Tsyvinski, A., Wu, X. (2022). Common risk factors in cryptocurrency. *Journal of Finance*, 77(2), 1133–1177.
- ▣ Markowitz, H. (1952). Portfolio selection. *Journal of Finance*, 7(1), 77–91.
- ▣ McLean, R.D., Pontiff, J. (2016). Does academic research destroy stock return predictability?. *Journal of Finance*, 71(1), 5–32.

References (6/7)

- ▣ Newey, W.K., West, K.D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.
- ▣ Onatski, A. (2010). Determining the number of factors from empirical distribution of eigenvalues. *Review of Economics and Statistics*, 92(4), 1004–1016.
- ▣ Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). Are cryptos becoming alternative assets?. *The European Journal of Finance*, 29(10), 1064–1105.
- ▣ Pesaran, M.H., Yamagata, T. (2024). Testing for alpha in linear factor pricing models with a large number of securities. *Journal of Financial Econometrics*, 22(2), 407–460.
- ▣ Politis, D.N., Romano, J.P. (1994). The stationary bootstrap. *Journal of the American Statistical Association*, 89(428), 1303–1313.
- ▣ Roll, R. (1977). A critique of the asset pricing theory's tests. Part I. *Journal of Financial Economics*, 4(2), 129–176.
- ▣ Ross, S.A. (1976). The arbitrage theory of capital asset pricing. *Journal of Economic Theory*, 13(3), 341–360.
- ▣ Shanken, J. (1992). On the estimation of beta-pricing models. *Review of Financial Studies*, 5(1), 1–33.
- ▣ Sharpe, W.F. (1964). Capital asset prices: a theory of market equilibrium under conditions of risk. *Journal of Finance*, 19(3), 425–442.

References (7/7)

- ▣ Vasicek, O.A. (1973). A note on using cross-sectional information in Bayesian estimation of security betas. *Journal of Finance*, 28(5), 1233–1239.
- ▣ Data: Kenneth R. French Data Library, mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.